

Event-Based UAV Detection via Spatiotemporal Voxel Clustering

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Abstract. We propose EV-SVC (Event-based Spatiotemporal Voxel Clustering), a learning-free drone detection pipeline for high-definition (HD) event camera streams. While learning-based methods in EV-UAV achieve strong performance on low-resolution event data, applying them to HD resolution (1280×720) causes GPU and system memory overflow due to the massive scale of HD event streams. EV-SVC captures the spatiotemporal continuity of drone trajectories through rule-based voxel clustering without any training, combining per-cell background subtraction, 3D connected component analysis, and adaptive filtering with age-based temporal continuity. We evaluate on the FRED dataset, a 7-hour HD outdoor benchmark including rain and nighttime conditions, using event-level metrics consistent with EV-UAV. Our method achieves $P_d = 66.5\%$ and $F_a = 0.58 \times 10^{-4}$, providing quantitative evaluation of learning-free drone detection on HD outdoor event camera data.

Keywords: Event camera · UAV detection · Spatiotemporal clustering · Learning-free · FRED · EV-UAV

1 Introduction

Detecting drones with conventional RGB cameras is inherently difficult under challenging real-world conditions. Motion blur degrades image quality at high flight speeds, limited dynamic range causes failures under backlight or nighttime illumination, and fixed frame rates (30–60 Hz) introduce latency that is incompatible with the rapid response times required for drone detection. Event cameras address these limitations through asynchronous per-pixel brightness change detection, offering microsecond temporal resolution and a dynamic range exceeding 120 dB [3].

Recent work has begun to exploit event cameras for UAV detection. The EV-UAV benchmark [2] establishes evaluation protocols and learning-based baselines for event-based drone detection, introducing the spatiotemporal continuity insight that drone trajectories form continuous curves in (x, y, t) space while static backgrounds form surfaces parallel to the t -axis. The FRED dataset [5] provides over 7 hours of HD outdoor event and RGB recordings (1280×720) under diverse conditions including rain and nighttime, enabling evaluation at realistic scale. On the learning-free side, Mandula *et al.* [6] demonstrate event-based UAV detection using dynamic vision sensors, and Magrini *et al.* [4] initially

pursue a learning-free approach but ultimately adopt a learning-based detector. Prior learning-free approaches on event-based UAV detection have not provided quantitative evaluation on HD outdoor data at the scale of FRED.

In this work, we address this gap. Given the memory constraints of applying learning-based methods to HD event streams, we propose EV-SVC (Event-based Spatiotemporal Voxel Clustering), a learning-free pipeline that operationalizes the spatiotemporal continuity principle through rule-based voxel clustering, without any training data or learned parameters.

Our contributions are as follows:

- We present quantitative evaluation of learning-free UAV detection on HD outdoor event camera data using the FRED dataset, with event-level metrics consistent with EV-UAV.
- We propose EV-SVC, combining per-cell background subtraction, 3D connected component analysis, adaptive displacement filtering, and age-based temporal continuity.
- We provide the first event-level P_d/F_a quantification of a learning-free method on HD FRED, without any training.

2 Related Work

Event-Based Object Detection and Tracking. Event cameras have been applied to a range of detection and tracking tasks owing to their microsecond temporal resolution and high dynamic range [3]. Asynchronous approaches process events individually as they arrive, maintaining probabilistic object models updated in real time; Wang *et al.* [7] propose an asynchronous blob tracker that maintains Gaussian blob models via extended Kalman filtering, demonstrating low-latency tracking of fast-moving objects. These methods target general moving objects and do not address drone-specific detection in HD outdoor environments.

Event-Based UAV and Drone Detection. The EV-UAV benchmark [2] introduces the first large-scale event camera dataset for drone detection with per-event annotations, and proposes EV-SpSegNet, which segments drone trajectories as continuous curves in (x, y, t) point cloud space using sparse 3D convolutions. A key observation motivating our work is that moving objects form continuous curves in (x, y, t) space while static backgrounds form surfaces parallel to the t -axis. The FRED dataset [5] provides over 7 hours of synchronized HD event and RGB recordings (1280×720) under diverse outdoor conditions including rain and nighttime, with learning-based baselines (YOLO, RT-DETR, Faster-RCNN) evaluated using mAP; no learning-free method is evaluated on FRED. Mandula *et al.* [6] detect UAVs using dynamic vision sensors with a spatter tracking approach, and Magrini *et al.* [4] initially pursue a learning-free strategy but ultimately adopt a learning-based multimodal detector. We similarly find that purely geometric criteria cannot achieve drone-specific discrimination, and we quantify this limitation on HD outdoor data. Spatiotemporal

clustering methods such as ST-DBSCAN [1] perform density-based clustering directly on raw (x, y, t) event coordinates; our approach approximates this on a discrete voxel grid, trading spatial precision for computational efficiency.

3 Method

EV-SVC processes a stream of events $\mathcal{E} = \{e_i = (x_i, y_i, t_i, p_i)\}$ from an HD event camera and outputs bounding box detections at 33 ms intervals, following the evaluation protocol of FRED [5]. The pipeline consists of six stages described below.

3.1 Stage 0: Baseline Initialization

The sensor plane is partitioned into non-overlapping cells of size $G \times G$ pixels. For each cell c , we maintain a baseline $B_c^{(t)}$ tracking its typical event activity via exponential moving average:

$$B_c^{(t)} = (1 - \alpha) B_c^{(t-1)} + \alpha F_c^{(t)}, \quad (1)$$

where $F_c^{(t)}$ is the event count in cell c at window t and $\alpha = 0.2$. The baseline is initialized over K_{warm} warm-up windows before detections are produced.

3.2 Stage 1: Background Subtraction

Events in cells where $F_c^{(t)} > B_c^{(t)}$ are retained as *residual events*, representing activity exceeding the estimated background level. This suppresses persistently active regions such as vegetation and buildings.

3.3 Stage 2: Voxel Grid Construction

Residual events from the most recent W_{buf} windows are accumulated into a 3D binary voxel grid of size $H_g \times W_g \times T_c$, with spatial resolution G px per cell and temporal resolution 1 ms per slice. A voxel is active if at least one residual event falls within it.

3.4 Stage 3: 3D Connected Component Analysis

We apply 6-connectivity connected component labeling to the constructed voxel grid, considering only axis-aligned neighbors $(\pm x, \pm y, \pm t)$. Each resulting component $\mathcal{C}_i$ represents a spatiotemporally contiguous cluster of residual activity.

3.5 Stage 4: Component Filtering

For each component $\mathcal{C}_i$, we compute its spatial extent (w_i, h_i) in pixels, its duration Δt_i in milliseconds, and its intra-component displacement:

$$d_i = \|\bar{c}_{t_{\max}} - \bar{c}_{t_{\min}}\|_2 \cdot p, \quad (2)$$

where $\bar{c}_{t_{\min}}$ and $\bar{c}_{t_{\max}}$ are the centroids of voxel cells at the earliest and latest active time slices, and $p = 5$ px is the grid pitch.

To accommodate distant drones whose pixel-space displacement is small due to perspective foreshortening, we apply an adaptive displacement threshold:

$$\delta_{\text{disp}} = \max(1, \min(w_i, h_i) \cdot 0.05). \quad (3)$$

A component is accepted as a candidate if:

$$\Delta t_i \geq \tau_{\text{dur}} \wedge d_i \geq \delta_{\text{disp}} \wedge s_{\min} \leq \min(w_i, h_i) \wedge \max(w_i, h_i) \leq s_{\max}, \quad (4)$$

where τ_{dur} , $s_{\min}$, and $s_{\max}$ are defined in Sec. 4.

3.6 Stage 5: Age-Based Temporal Continuity

Candidate components are matched across consecutive windows by centroid distance. A matched candidate increments its *age* counter a ; an unmatched candidate increments a *miss* counter. A candidate is discarded when its miss count exceeds M_{miss} . A candidate is promoted to a confirmed detection only when:

$$a \geq \tau_{\text{age}}, \quad (5)$$

where τ_{age} is defined in Sec. 4. This rejects transient false positives from rain and wind-driven vegetation while retaining genuine drone trajectories that persist across multiple windows.

4 Experiments

4.1 Dataset

We evaluate on the FRED dataset [5], comprising 231 sequences with over 7 hours of synchronized HD event and RGB recordings (Prophesee EVK4, 1280×720, >120 dB). The dataset covers five drone models under diverse conditions including rain, nighttime, and airborne distractors. Since EV-SVC requires no training, we evaluate on all 231 sequences across both the canonical train and test splits.

4.2 Evaluation Protocol

We adopt the event-level evaluation protocol of EV-UAV [2]. Ground truth bounding boxes are extended along the time axis to generate per-event binary target labels: an event (x_i, y_i, t_i) is labeled $\text{target} = 1$ if it falls within any GT bounding box interval, and $\text{target} = 0$ otherwise. An event is labeled $\text{predicted} = 1$ if its (x, y) falls within any confirmed bounding box in the corresponding window.

The detection rate is computed at the object level, where an object is defined as a $(\text{drone_id}, 33 \text{ ms window})$ pair, following the evaluation interval $\tau = 33 \text{ ms}$ set by FRED [5] to match the RGB camera frame rate:

$$P_d = \frac{N_{\text{detected}}}{N_{\text{total}}}, \quad (6)$$

where an object is considered detected if the fraction of its target events covered by predicted regions exceeds $\theta = 0.5$. The false alarm rate is:

$$F_a = \frac{N_{\text{false blobs}}}{N_{\text{frames}} \times H \times W}, \quad (7)$$

where false alarm blobs are 2D connected components formed by background events incorrectly labeled as predicted, and $H \times W = 720 \times 1280$ corresponds to the FRED sensor resolution.

We verify P_d via a null test: randomizing GT bounding box positions reduces P_d from 78.7% to 4.0% on seq 82, confirming that the metric reflects genuine detection signal. While EV-UAV assigns predicted labels via per-event neural network confidence scores, we use confirmed bounding box regions as a proxy, as EV-SVC does not produce per-event scores.

4.3 Implementation Details

EV-SVC is implemented in Python using `scipy.ndimage.label` for 3D connected component analysis. Parameters: grid pitch $G = 5 \text{ px}$, temporal resolution 1 ms , buffer $W_{\text{buf}} = 2 \text{ windows}$ (66 ms), $\alpha = 0.2$, warm-up $K_{\text{warm}} = 100 \text{ windows}$ (3.3 s), $\tau_{\text{dur}} = 33 \text{ ms}$, $s_{\text{min}} = 5 \text{ px}$, $s_{\text{max}} = 200 \text{ px}$, $\tau_{\text{age}} = 4 \text{ windows}$, $M_{\text{miss}} = 4 \text{ windows}$.

4.4 Results

Tab. 1 presents results on all 231 FRED sequences. EV-SVC achieves $P_d = 66.5\%$ and $F_a = 0.58 \times 10^{-4}$. EV-UAV results are included for contextual reference only, as they are evaluated on a different dataset (346×260 resolution) and are not directly comparable.

Table 1: Detection results. EV-UAV methods [2] are shown for contextual reference only (different dataset and resolution).

Method	Type	P_d (%) $\uparrow$	F_a ($\times 10^{-4}$) $\downarrow$
SSD [2]	Learning	26.31	486.63
RVT [2]	Learning	60.35	55.68
KPConv [2]	Learning	68.59	16.32
EV-SpSegNet [2]	Learning	77.53	1.63
EV-SVC (ours)	Learning-free	66.5	0.58

Table 2: Ablation study on seq 82.

Configuration	P_d (%) $\uparrow$	F_a ($\times 10^{-4}$) $\downarrow$
Baseline	77.51	4.06
+ B-2 (adaptive disp)	78.69	4.06
+ B-3 ($\tau_{\text{age}} = 0$)	79.92	4.43
+ B-3 ($\tau_{\text{age}} = 4$)	76.03	2.70
+ B-3 ($\tau_{\text{age}} = 6$)	73.62	2.14

4.5 Ablation Study

Tab. 2 shows component contributions on seq 82. B-2 improves P_d by recovering small and distant drones whose pixel-space displacement falls below the fixed threshold. B-3 reduces F_a by suppressing transient false positives, with the trade-off between P_d and F_a controlled by τ_{age} .

5 Analysis

5.1 Performance Variation Across Sequences

EV-SVC exhibits substantial variation across the 231 evaluated sequences, with P_d ranging from 13.0% (seq 67) to 97.3% (seq 193) and F_a ranging from near zero (seq 111, 0.0001×10^{-4}) to 21.21×10^{-4} (seq 84). Two primary factors drive this variation.

Drone size and voxel granularity. Sequences with high P_d feature drones occupying a large image-plane area, producing dense residual events that form well-connected voxel components. Sequences with low P_d (e.g., seq 67) feature distant drones that appear as only a few pixels, often within a single voxel cell ($G = 5$ px). At this scale, spatiotemporal components are too sparse to satisfy the duration and displacement thresholds, resulting in missed detections. Reducing G to 2–3 px would address this but increases the voxel grid size by $6\times$ and $2.7\times$ respectively, significantly impacting processing speed. This represents a fundamental resolution–efficiency trade-off inherent to fixed-resolution voxelization.

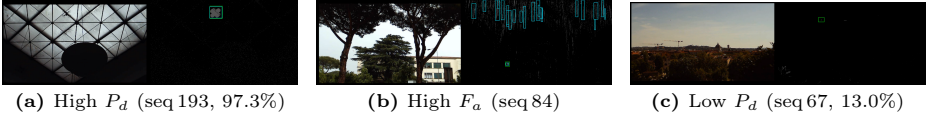


Fig. 1: Qualitative results. (a) High- P_d case: drone clearly detected (green box). (b) High- F_a case: heavy rain produces numerous false alarms (cyan). (c) Low- P_d case: distant drone is sub-voxel scale, missed detection. Each panel shows the RGB frame (left) and event detections (right).

Rain and environmental disturbance. Stage-level analysis on seq 84 (heavy rain) reveals the mechanism: Stage 3 produces on average 13,004 raw components per window, of which Stage 4 passes only 17.1 per window (0.13%). Despite this high rejection rate, the sheer volume of rain-induced components means that the surviving candidates per window are predominantly rain-generated false positives, directly causing the elevated $F_a = 21.21 \times 10^{-4}$. Rain droplets generate spatiotemporal trajectories in (x, y, t) that satisfy displacement and size criteria, making them geometrically indistinguishable from drone motion under learning-free criteria.

Fig. 1 illustrates representative cases.

Qualitative visualization videos for challenging scenarios (rain, nighttime, and strong illumination) are provided in the supplementary material.

5.2 Parameter Sensitivity

We analyze the sensitivity of two key parameters on seq 82.

Age threshold τ_{age} . As shown in Tab. 2, increasing τ_{age} reduces F_a by suppressing transient false positives, at the cost of reduced P_d as genuine drone detections require longer temporal persistence to be confirmed. This trade-off allows the operating point to be adjusted depending on the application’s tolerance for false alarms.

Buffer size W_{buf} . Extending the buffer from 3 windows (100 ms) to 6 windows (200 ms) increases F_a from 4.06×10^{-4} to 7.77×10^{-4} without a corresponding improvement in P_d (77.5% vs. 77.8%). A longer buffer accumulates more false positive activations from environmental sources without recovering additional drone detections. The default 66 ms buffer provides a better balance.

Both analyses are conducted on seq 82 only; generalization across all sequences is left for future work.

6 Conclusion

We presented EV-SVC, a learning-free drone detection pipeline for HD event camera streams that operationalizes the spatiotemporal continuity principle through rule-based voxel clustering. Without any training data or learned parameters, EV-SVC achieves $P_d = 66.5\%$ and $F_a = 0.58 \times 10^{-4}$ on the FRED

dataset, providing quantitative evaluation of learning-free drone detection on HD outdoor event camera data.

The primary limitations of EV-SVC are its fixed voxel resolution ($G = 5$ px), which causes missed detections for distant drones that occupy only a single voxel cell, and the absence of drone-specific discrimination criteria, which causes physically moving non-drone objects such as vehicles and rain to generate false positives that are geometrically indistinguishable from drone trajectories under learning-free criteria.

Future work includes two directions. First, finer voxel resolution ($G = 2-3$ px) to improve detection of small and distant drones, combined with sparse voxel representations to manage the associated increase in computational cost. Second, multimodal fusion in which EV-SVC acts as a pre-filter that narrows the spatial and temporal regions of interest, reducing the volume of RGB or acoustic data that must be processed for drone-specific verification, without relying on learned models for either modality.

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